Is That for Me? Addressee-Aware Turn-Keeping for Open Full-Duplex Speech Models under Same-Voice Side-Talk

Team Member 1 and Team Member 2 and Team Member 3 and Team Member 4
ECE 6256 Conversational AI (Fall 2026), School of ECE, Georgia Institute of Technology
{member1,member2,member3,member4}@gatech.edu

Abstract

Open full-duplex speech models treat every voice in the user channel as speech meant for them, so on three 2026 benchmarks they answer when the user turns to a third person. The shipped fix (always-on "proactive audio" in Gemini 3.8 Live) has no published method, and open remedies gate on speaker identity, which cannot help when the side-talk is in the user's own voice. We propose to measure how much identity gating solves, build a de-shortcut same-voice side-talk test set with voice cloning, LoRA-fine-tune Moshi on cue-controlled synthetic side-talk, and ablate lexical against acoustic cues.

1 Problem and gap

A full-duplex model must decide, frame by frame, whether incoming speech is for it. The 2026 survey of full-duplex systems (Lu et al., 2026) lists "third-party / not-directed speech" as an ontology cell that speaker-embedding and relevance gates target but have "not shown solved". The failure is measured, not fixed. On Full-Duplex-Bench (FDB) v1.5 (Lin et al., 2025), when the user talks to someone else, GPT-4o Realtime responds on 91% of clips and Freeze-Omni on 58%; Moshi responds on 20% and is unclassifiable on another 59%, and the authors conclude that systems "must strengthen addressee detection". The HumDial challenge (Wang et al., 2026) and MSI-Bench (Xiong et al., 2026) report the same failure: "models often respond even when no one addressed them". Gemini 3.8 Live keeps "proactive audio" (the model "can proactively decide not to respond if the content is not relevant") permanently enabled (Google, 2026) and Realtime-Venus (Ant Group, 2026) reports 88% correct ignoring on this cell; neither releases weights or a method.

Every open remedy gates on *who* is speaking: FireRedChat's personalised VAD (Chen et al., 2025) and IRAF's ECAPA-embedding gate inside an end-to-end model (Zhong et al., 2026). Kyutai's interactivity alignment (Ohashi et al., 2026) has no addressee axis. Yet FDB v1.5 deliberately separates two cells: *bkg_speech* is a different voice at -15 dB, whereas *talking_other* is the *user's own voice* at -8 dB with far-field reflections, saying something meant for a third person. Identity gating can only address the first. No public model has been trained on same-voice side-talk and no corpus labels it at scale (HumDial's training set has zero such items).

2 Research questions and hypotheses

RQ1 Which part of the not-addressed cell does identity gating solve? *H1:* a verification gate mutes most different-voice speech but passes most same-voice side-talk. **RQ2** Can a Moshi-class model learn to keep its turn under same-voice side-talk while still yielding to genuine interruptions? *H2:* LoRA on cue-controlled synthetic side-talk lowers the side-talk respond rate by ≥ 20 points at ≤ 5 points more missed interruptions. **RQ3** Which cue does it learn? *H3:* with acoustic cues (level drop, far-field colouring) present, the gain survives removing the lexical cue but not the acoustic one.

3 Proposed system

Base and baselines. Moshiko-7B (Défossez et al., 2024) (CC-BY 4.0, official `moshi-finetune` LoRA) is the fine-tuning target; PersonaPlex-7B (Roy et al., 2026) and Freeze-Omni are inference baselines. **G0 identity gate (no training).** ECAPA cosine between each 0.5–1 s user-channel window and an enrollment from the user's earlier speech; windows below a held-out-tuned threshold are zeroed before the model hears them. It ports pVAD/IRAF to an open model (RQ1). **Data pipeline.** Behavior-SD (Lee et al., 2025) (108k synthetic full-duplex dialogues, 2,164 h, CC-BY 4.0) supplies base dialogues. An LLM writes side-talk

addressed to a named third party (with and without a vocative) and genuine interruptions; Chatterbox voice cloning renders them in the user's voice; they enter the user channel under three acoustic conditions: FDB-matched far-field (-8 dB, high-shelf cut, reflections), near-field (lexical cue only) and content-neutral far-field (acoustic cue only). Different-voice background speech (LibriSpeech, 0–15 dB SNR) is the control cell. **Training target.** Moshi keeps producing its utterance (no PAD, no silence) through non-addressed speech and yields on addressed interruptions; LoRA rank 128, about 2,000 steps, one H100 (≈40 GB).

4 Experimental design

Test sets. (i) FDB v1.5: talking_other, bkg_speech, backchannel, interruption (498 clips). (ii) HumDial-FDBench test set (600 third-party and 200 talking-to-others items, human actors). (iii) Our de-shortcut set: same vs. different voice $\times$ acoustic cue $\times$ vocative, on unseen voices, with far-field genuine interruptions and "Hey Moshi, ..." vocatives aimed at the system. (iv) A small human-recorded set (50–100 clips). **Metrics** (conditions: Moshi, +G0, +LoRA, +G0+LoRA, PersonaPlex, Freeze-Omni)**.** Respond rate on non-addressed speech (FDB labels plus VAD-based measures, since Moshi's labels are mostly "unknown"); response rate and stop latency on genuine interruptions; spurious-onset rate; FDB content-quality judge; per-frame gate latency within the 80 ms frame budget. **Estimands and analysis.** Primary: change in side-talk respond rate (LoRA vs. base) with the interruption miss rate held within 5 points. Secondary: G0's split between different-voice and same-voice speech. Paired bootstrap CIs by clip; three LoRA seeds; the 2×2 cue ablation on set (iii) answers RQ3.

5 Pilot evidence

On FDB v1.5 (498 clips, ECAPA) the identity cue behaves as H1 predicts (Table 1): the gate separates background speech from real interruptions almost perfectly (AUC 0.9998) yet accepts 88–92% of same-voice side-talk at thresholds that keep 95% of the user's own speech. The pilot also exposed two shortcuts: 99 of 100 side-talk clips open with a vocative ("Dad, ..."), so a comma-within-three-words rule flags 99% at 3% false positives, and all side-talk is far-field while all interruptions are near-field. Chatterbox clones of the user voice pass as the same voice and render at 5–7 s per utterance on an RTX 3070, so the de-shortcut set is an overnight job.

FDB v1.5 scenario	cos.	pass	muted
user_interruption (keep)	.49	.995	.13
talking_other (want mute)	.39	**.89**	**.27**
user_backchannel (keep)	.19	.52	.47
bkg_speech (mute)	.06	.01	.83

Table 1: Identity cue on FDB v1.5: median ECAPA cosine to the user; clips accepted as the user at the EER threshold; audio muted by the 0.5 s G0 gate.

6 Plan, roles and feasibility

Week 1 (to 11/12): PACE setup, three-model baselines on FDB v1.5 and HumDial, G0, LoRA smoke test, synthesis v0. Milestone 11/19: baseline table with the G0 split, first LoRA. Weeks 3–4: LoRA sweep over cue conditions and data size, de-shortcut and human sets, latency. Poster 12/3; paper and code 12/8. One member each on evaluation and baselines, synthesis and test sets, G0 and analysis, LoRA. Inference needs ≈24 GB (L40S) and LoRA one H100, inside the PACE-ICE allocation.

7 Risks and fallback

(1) The model may learn "quieter means ignore" rather than addressee semantics; the cue ablation turns this into a finding. (2) `moshi-finetune` is validated only on Moshiko; a week-1 smoke test gates the rest. (3) Synthetic side-talk may not transfer to real rooms; HumDial's human recordings and our recorded set bound the gap. If LoRA fails, the deliverable is a measurement paper: three models $\times$ three benchmarks $\times$ four cells, the identity-gate split and the de-shortcut analysis.

Plan B (piloted, text-only). Provenance-tracked record maintenance on ConFormBench (Veigel and Mayer, 2026), with the model's own output as the next turn's prior state. Our closed-loop pilot found that errors contradicted by visible history are always repaired, but fabricated values (5/7) and ambiguous readings (5/7) persist, even through the shipped verification stage, and a fact outside a 3-turn history window stays wrong 8/8 times. The method keeps per-field evidence spans and ambiguity flags and re-verifies only unlicensed or flagged fields, extending Correctable-DST (Xie et al., 2022); about 100 gold trajectories, no GPU.

References

Ant Group. 2026. Realtime-Venus: A full-duplex interaction system with asynchronous delegation. *arXiv preprint arXiv:2609.13814*.

Junjie Chen, Yao Hu, Junjie Li, Kangyue Li, Kun Liu, Wenpeng Li, Xu Li, Ziyuan Li, et al. 2025. FireRedChat: A pluggable, full-duplex voice interaction system with cascaded and semi-cascaded implementations. *arXiv preprint arXiv:2509.06502*.

Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. 2024. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*.

Google. 2026. Gemini 3.8 live: model page and live API documentation. `https://ai.google.dev/gemini-api/docs/models/gemini-3.8-live`. Accessed 2026-11-03.

Sehun Lee, Kang-wook Kim, and Gunhee Kim. 2025. Behavior-SD: Behaviorally aware spoken dialogue generation with large language models. In *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*.

Guan-Ting Lin, Shih-Yun Shan Kuan, Qirui Wang, Jiachen Lian, Tingle Li, Shinji Watanabe, and Hung-yi Lee. 2025. Full-Duplex-Bench v1.5: Evaluating overlap handling for full-duplex speech models. *arXiv preprint arXiv:2507.23159*.

Jingyu Lu, Yuhan Wang, Jianming Luo, Yifu Chen, Tianle Liang, Shengpeng Ji, Ziyue Jiang, Xiaoda Yang, et al. 2026. A survey of full-duplex spoken dialogue systems: Architectural hierarchy, interaction ontology, and decision state machine. *arXiv preprint arXiv:2606.19453*.

Atsumoto Ohashi, Neil Zeghidour, Alexandre Défossez, and Eugene Kharitonov. 2026. Multi-faceted interactivity alignment in full-duplex speech models. *arXiv preprint arXiv:2606.11167*.

Rajarshi Roy, Jonathan Raiman, Sang-gil Lee, Teodor-Dumitru Ene, Robert Kirby, Sungwon Kim, Jaehyeon Kim, and Bryan Catanzaro. 2026. PersonaPlex: Voice and role control for full duplex conversational speech models. *arXiv preprint arXiv:2602.06053*.

Timothy Veigel and Sven Mayer. 2026. ConFormBench: Evaluating evidence-supported record maintenance in conversational form filling. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*.

Chengyou Wang, Hongfei Xue, Guojian Li, Zhixian Zhao, Shuiyuan Wang, Shuai Wang, Xin Xu, Hui Bu, and Lei Xie. 2026. Full-duplex interaction in spoken dialogue systems: A comprehensive study from the ICASSP 2026 HumDial challenge. *arXiv preprint arXiv:2604.21406*.

Hongyan Xie, Haoxiang Su, Shuangyong Song, Hao Huang, Bo Zou, Kun Deng, Jianghua Lin, Zhihui Zhang, and Xiaodong He. 2022. Correctable-DST: Mitigating historical context mismatch between training and inference for improved dialogue state tracking. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 876–889.

Chenxu Xiong, Dongming Shen, Yuzhi Tang, Wentao Ma, Mu Li, and Alex Smola. 2026. MSI-Bench: Evaluating multi-speaker voice interaction for collaborative AI agents. *arXiv preprint arXiv:2609.24812*.

Tao Zhong, Jiajun Deng, Nikita Kuzmin, Yinke Zhu, Tianxiang Cao, Tristan Tsoi, Zhili Tan, Simon Lui, et al. 2026. IRAF: Interference-resilient adaptive fusion for noise-robust end-to-end full-duplex spoken dialogue systems. *arXiv preprint arXiv:2606.06559*.